

Task Coordinates for Few-Shot Diffusion Adaptation: Results Against the Experimental Plan

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Paper under double-blind review

17 September 2026

Abstract

Meta-Diffusion adapts a frozen denoising diffusion probabilistic model to a new data domain by fitting a low-dimensional task coordinate rather than by updating network weights. This report gives the results of the current experimental stage, block by block against the governing plan. The plan fixes the criterion in advance: the main claim holds only if the correct source task together with a learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling. That criterion is met when target data are scarce, and only then. Six of the ten deliverables are complete; four are not, and each is named with its reason.

1 CRITERION AND NOTATION

Criterion. The main claim is supported only if the correct source task *plus* the learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling, in the low-target-data regime.

Notation. A set encoder maps a set of images to a task coordinate $z \in \mathbb{R}^k$. Writing z_S for the coordinate encoded from source-domain data and c for a descriptor of the source-to-target relation, the transport map predicts the target coordinate $\tilde{z}_T = z_S + \Delta_\gamma(z_S, c)$, taking no target image as input. M_S is the number of source images the encoder sees; K_T is the number of target images available at adaptation time. Feature-wise linear modulation, the standard alternative to an additive basis, scales and shifts intermediate feature maps as a function of the coordinate; it is called the modulation arm below.

Metric. Distances are the sliced Wasserstein distance to the real target set in pixel space, which the plan names as headline; lower is better. Every comparison is a paired difference over episodes with a 95% confidence interval, and an asterisk marks an interval excluding zero.

Configuration. One checkpoint of the three-transformation CIFAR-100 domain-shift family — Gaussian blur, additive Gaussian noise, contrast reduction at severity 3 — a U-Net backbone of 128 base channels, coordinate dimension $k = 16$, 50,000 training steps read with exponentially moving averaged weights. Adaptation uses 25 refinement steps at rate 10^{-2} . Generation uses the deterministic implicit sampler, 50 reverse steps, 96 samples per cell, 60 episodes per cell.

2 STAGE 0 — PRECONDITIONS

3 STAGE A — CIFAR-100, GENERATION-LEVEL EVIDENCE

3.1 A1 — Source-dependence controls

180 episodes, twenty classes by three transformations by $K_T \in \{1, 5, 20\}$, nine conditions.

Precondition	Status
0.1 Integrate the three branches	Not done. Both modulation-arm source files are absent from this tree. Blocks A2.
0.2 Verify the split file	Done; hash matches the fixed value.
0.3 Modulation checkpoint	Not done. Absent from this machine. Also blocks A2.
0.4 Evaluation primitives	Four of five written. The kernel Inception distance was not implemented.
0.5 Output directory	Done.

Full method against	$K_T = 1$	$K_T = 5$	$K_T = 20$
Mean z_S	$+0.0301 \pm 0.0072^*$	$+0.0346 \pm 0.0085^*$	$+0.0338 \pm 0.0080^*$
Shuffled z_S	$+0.0416 \pm 0.0125^*$	$+0.0453 \pm 0.0149^*$	$+0.0436 \pm 0.0144^*$
Relation-only	$+0.0254 \pm 0.0078^*$	$+0.0285 \pm 0.0088^*$	$+0.0295 \pm 0.0087^*$
Target-only	$+0.018 \pm 0.009^*$	$+0.011 \pm 0.007^*$	-0.003 ± 0.007
Oracle (negative = behind)	-0.026^*	-0.027^*	-0.017^*

Table 1: A1. Paired differences; positive favours the full method.

Conclusion. The method beats all three source controls at every budget. It beats relation-only transfer and target-only modelling simultaneously at one and five target images, which is what the criterion requires, and the advantage over target-only modelling is gone by twenty. **The criterion is met in the low-target-data regime and is bounded there:** source information is worth having when target evidence is scarce, and its value decays as target evidence accumulates. A gap to the oracle remains at every budget.

3.2 A2 — Additive basis against the modulation arm

Not run. Preconditions 0.1 and 0.3 are both unmet: the modulation implementation is on another branch and its checkpoint is not on this machine. Both are repository operations, not experiments. The arm exists and has been measured on the denoising loss; only the generation-level comparison is outstanding.

3.3 A3 — Target-scarcity sweep

300 episodes at $K_T \in \{1, 2, 5, 10, 20\}$.

Full method against	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Reuse z_S	$+1.140^*$	$+0.915^*$	$+0.526^*$	$+0.317^*$	$+0.229^*$
Half-width	0.108	0.105	0.099	0.067	0.059
Target-only	$+0.061^*$	$+0.038$	$+0.005$	$+0.018$	$+0.008$
Half-width	0.051	0.052	0.036	0.045	0.041

Table 2: A3. Paired differences in transformation-statistic space, the instrument this sweep was run with.

Conclusion. The method beats reusing the source coordinate at every budget, the advantage decaying monotonically by a factor of five and remaining clear of zero throughout. This is the largest effect in the stage. Against target-only modelling the advantage is confined to the smallest budget. **The plan asked for the crossing point to be named: there is no crossing, only a convergence** — the two become indistinguishable from two target images onward, and the method was never shown to fall behind.

3.4 A4 — Source-abundance sweep

Six source budgets from 16 to 300 images, one target image, 60 episodes each.

A4: source abundance at deployment, one checkpoint throughout

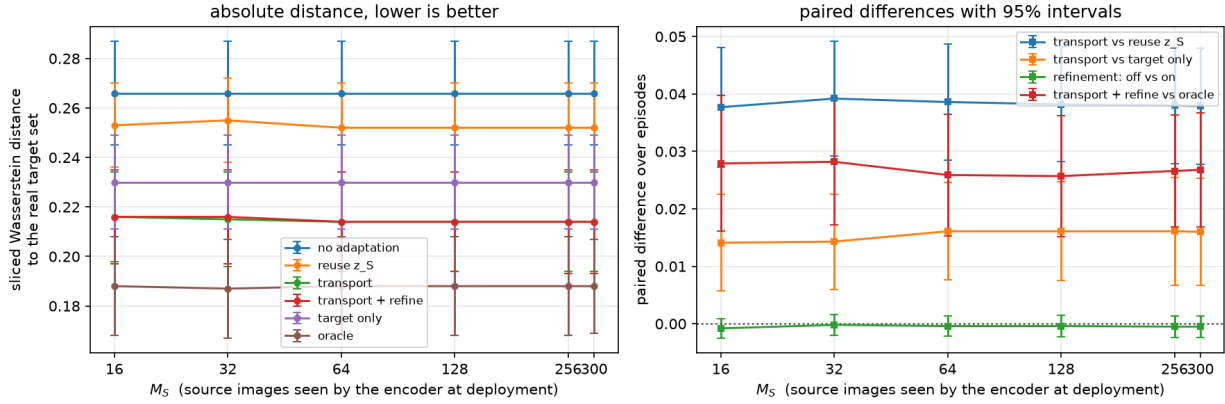


Figure 1: A4. Left: absolute distance per strategy. Right: paired differences with 95% intervals.

Conclusion. Across a nineteenfold range nothing moves: the comparison against reusing the source coordinate shifts by 0.0002, that against target-only modelling by 0.0019, that against the oracle by 0.0011, each smaller than the half-width of its own interval. The plan expected improvement followed by saturation. **What the sweep shows is saturation from the first point measured: sixteen source images already buy everything this transport map extracts from a source set.** The method is therefore cheap on the source side. The sweep varies source evidence available at deployment; the checkpoint was trained once, at $M_S = 64$.

3.5 A5 — Transport-structure ablation

Four maps trained separately under identical conditions, compared episode by episode over 180 episodes each.

Variant	Parameters	$K_T = 1$	$K_T = 5$	$K_T = 20$
Identity	0	$+0.0242 \pm 0.0080^*$	$+0.0256 \pm 0.0089^*$	$+0.0246 \pm 0.0087^*$
Constant relation vector	96	$+0.0082 \pm 0.0051^*$	$+0.0092 \pm 0.0052^*$	$+0.0109 \pm 0.0055^*$
Linear map	352	$+0.0022 \pm 0.0049$	$+0.0004 \pm 0.0048$	$+0.0036 \pm 0.0050$
Run-to-run floor	—	0.0076	0.0036	0.0135

Table 3: A5. Paired against the residual multilayer perceptron (6824 parameters); positive favours it. The floor is the spread across the same four runs of a strategy that never consults the transport map.

Conclusion. The transport map must be more than the identity, by a margin well above the floor at every budget. The constant relation vector’s margin sits inside the floor. **The linear map is indistinguishable from the residual network with nineteen times fewer parameters**, so the architecture can be simplified rather than defended.

4 STAGE B — GAUSSIAN-MIXTURE POSITIVE CONTROL

Five task families spanning a range of source-to-target distances, 32 tasks each.

Conclusion. The mechanism behaves as designed where the ground truth is analytic: the gain is entirely task-specific and the profile is minimised in the right place in all five families. Placed beside the same measurement on images, the well is 26.1% deep on mixtures and 1.6% deep on CIFAR-100, both minimised at the task’s own coordinate. **The mechanism on images is neither absent nor misdirected; it is shallow.** One observation is unexplained: the relatedness sweep does not move the task-specific term at all, although it was built to.

Task family	perturb 0.1	perturb 0.25	perturb 0.5	perturb 1.0	unrelated
Task-specific gain	+0.1539*	+0.1428*	+0.1536*	+0.1571*	+0.1533*
Profile minimum at	$s = 1$	$s = 1$	$s = 1$	$s = 1$	$s = 1$

Table 4: Stage B. The improvement from a task’s own coordinate over the coordinate shared by all tasks, and the location of the coordinate-loss minimum.

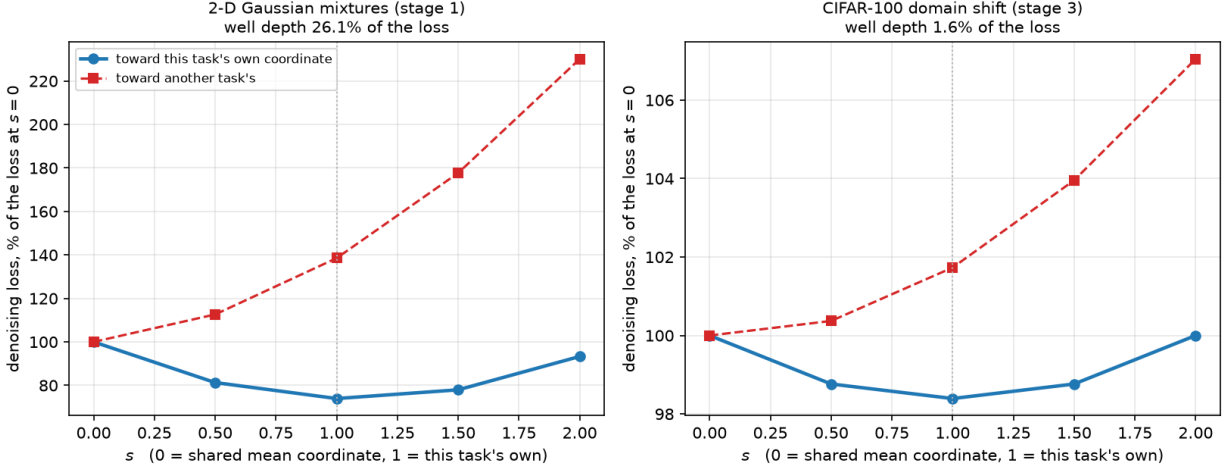


Figure 2: The coordinate-loss profile on Gaussian mixtures (left) and CIFAR-100 (right), plotted as a percentage of the loss at the shared mean coordinate so that the two are comparable.

5 STAGE C — DEMOGRAPHIC SCARCITY EXPERIMENT

5.1 C0 — Feasibility check (hard gate)

	Rows	Live images sampled
Archive 1	12,631 (76.2%)	0/30; domain fails to resolve
Archive 2	3,905 (23.6%)	30/30
Conditions above threshold, reachable subset		16
Of those, able to fund a 20-image support set		8
Conditions with the full dataset		83

Table 5: C0. Phototype grouping I–III against IV–VI, chosen from the observed counts.

Conclusion. The annotations are fully available and the images are not. Three quarters of the dataset sits behind a domain that no longer resolves, and recovering it depends on a release the dataset’s authors control. **The reachable subset supports eight usable conditions, which gives a five/two/one task split; one held-out test task cannot carry a paired interval.** It is therefore enough to build C1–C3 on, and not enough for the headline result of C4.

5.2 C1–C5 — Not run

	Item	Reason
C1	Domain module and splits	Not started. Buildable today on the reachable subset; not blocked.
C2	Frozen evaluation predictors	Not started; depends on C1.
C3	Optional sanity check	Not started; depends on C1 and C2.
C4	The scarcity sweep	Blocked by C0. The target-data equivalence statement the plan asks for needs a paired interval on held-out tasks, which eight conditions cannot provide.
C5	Downstream augmentation	Optional and conditional on C4.

6 STAGE D — FEW-SHOT FACE GENERATION

No new runs, as the plan directs. The existing negative result stands, reinterpreted as a task-family mismatch: the standard setup offers a handful of heterogeneous target domains rather than many instances of one reusable relation. A1 supports that reading, since the method’s advantage is concentrated where a reusable relation carries information a single target image does not.

7 DELIVERABLES

#	Deliverable	Status
1	Gaussian-mixture mechanism and calibration	Complete
2	Source-dependence controls	Complete
3	Additive basis against the modulation arm	Not run — preconditions 0.1 and 0.3 unmet
4	Target-scarcity curve, crossing named	Complete; there is no crossing
5	Source-abundance curve	Complete; saturated from the first point
6	Transport-structure ablation	Complete
7	One demographic scarcity experiment	Not run — blocked by C0
8	Paired 95% intervals for every headline	Complete
9	Downstream augmentation (optional)	Not run — depends on C4
10	Few-shot face stress test (optional)	Not run — excluded by the plan

8 CONCLUSION

The criterion fixed before the stage began is met, and met narrowly: the correct source task together with the learned relation beats both relation-only transfer and target-only modelling when target data are scarce, with the margin over target-only modelling gone by twenty target images. The positive control confirms the mechanism is sound where the ground truth is known and locates the difficulty on images as shallowness rather than absence. Two further results are usefully negative: source evidence saturates below the smallest budget measured, so the method is cheap on the source side, and the non-linear transport map is not distinguishable from a linear one, so the architecture can be simplified.

Three items would change what can be claimed. The unconstrained fine-tuning reference, which is implemented but was not run, would establish whether the remaining gap to the oracle belongs to the task or to the architecture. The branch integration of 0.1 unblocks deliverable 3 and is a repository operation. And the image release behind C0 is the only route to a task family in which a single target example does not already reveal the task — which, on the evidence of A1, is where this method has the most to offer.